

Lipid Chemistry

Definition, functions, classification, fatty acids, essential fatty acids and phospholipids

Core idea:

Lipids are a heterogeneous group of water-insoluble, organic-solvent-soluble substances. They are important as concentrated energy reserves, membrane components, insulating materials and metabolic regulators.

1. Definition of Lipids

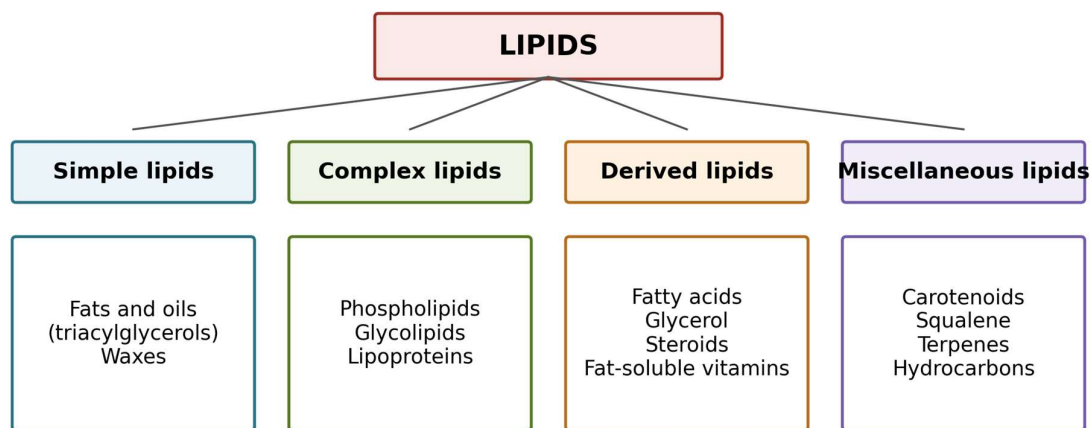
Lipids are organic substances that are relatively insoluble in water but soluble in organic solvents such as alcohol, ether and chloroform. They are actually or potentially related to fatty acids and are utilized by living cells.

Unlike carbohydrates, proteins and nucleic acids, lipids are not true polymers. Most lipids are relatively small molecules and are chemically diverse.

2. Functions of Lipids in the Body

Function	Explanation
Concentrated fuel reserve	Triacylglycerols stored in adipose tissue provide the body's major long-term energy reserve.
Membrane structure	Phospholipids and cholesterol form biological membranes and regulate membrane permeability.
Source of fat-soluble vitamins	Dietary lipids help supply and absorb vitamins A, D, E and K.
Metabolic regulation	Steroid hormones, prostaglandins and other lipid derivatives regulate many biochemical processes.
Protection and insulation	Fat protects internal organs, reduces heat loss and gives body contour and smooth appearance.
Transport of lipids	Lipoproteins transport triacylglycerols, cholesterol and phospholipids through blood.

3. Classification of Lipids with Examples



Based on Bloor classification used in the source text

Readable classification chart of lipids.

Class	Meaning	Examples
Simple lipids	Esters of fatty acids with alcohols.	Fats and oils (triacylglycerols), waxes
Complex lipids	Lipids containing fatty acids plus extra groups such as phosphate, carbohydrate, nitrogenous base or protein.	Phospholipids, glycolipids, lipoproteins
Derived lipids	Products obtained by hydrolysis of simple and complex lipids that still show lipid properties.	Fatty acids, glycerol, mono- and diacylglycerols, steroid hormones, fat-soluble vitamins
Miscellaneous lipids	Compounds having lipid-like properties but not fitting above groups.	Carotenoids, squalene, terpenes, hydrocarbons

Neutral lipids:

Uncharged lipids such as monoacylglycerols, diacylglycerols, triacylglycerols, cholesterol and cholesteryl esters are called neutral lipids.

4. Fatty Acids: Definition and General Features

Fatty acids are carboxylic acids with a hydrocarbon side chain. They are the simplest form of lipids and major constituents of many complex lipids.

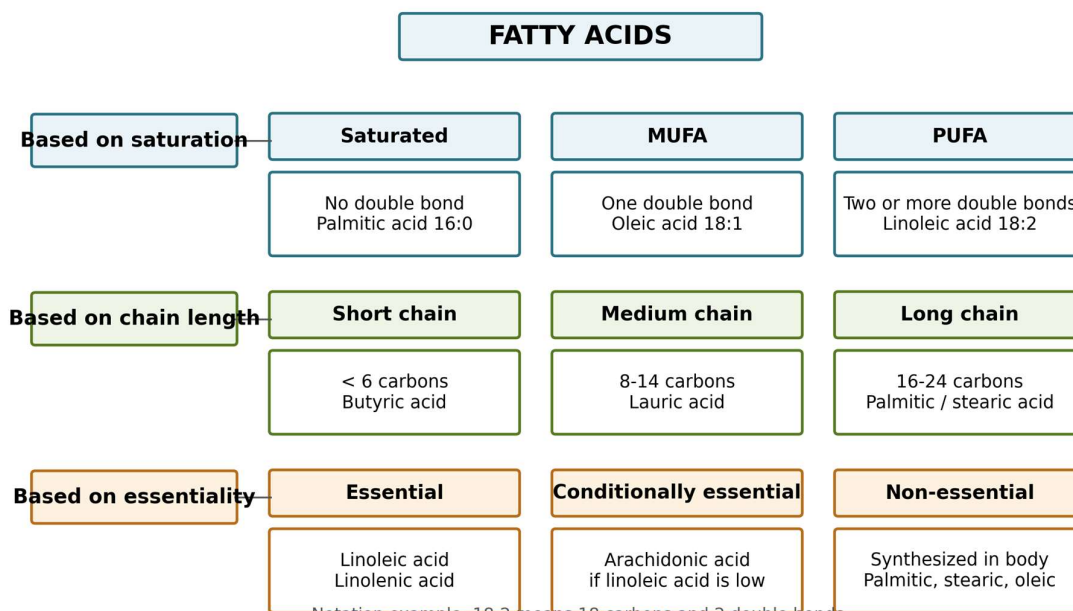
Most naturally occurring fatty acids have an even number of carbon atoms, usually between 14 and 20 carbons. This is because fatty acid synthesis occurs mainly by sequential addition of two-carbon units.

A fatty acid has two ends: a carboxyl end (-COOH/-COO-) and a methyl end (-CH₃). The long hydrocarbon chain is hydrophobic, while the carboxyl group is polar.

Shorthand notation	Example	Meaning
16:0	Palmitic acid	16 carbons and no double bond
18:1; 9	Oleic acid	18 carbons, one double bond at carbon 9
18:2; 9,12	Linoleic acid	18 carbons, two double bonds at carbons 9 and 12

Shorthand notation	Example	Meaning
20:4; 5,8,11,14	Arachidonic acid	20 carbons, four double bonds at carbons 5, 8, 11 and 14

5. Classification of Fatty Acids with Examples



Notation example: 18:2 means 18 carbons and 2 double bonds.

Readable summary of fatty acid classification.

A. Based on saturation

Type	Definition	Examples
Saturated fatty acids	No double bonds in the hydrocarbon chain.	Palmitic acid (16:0), Stearic acid (18:0)
Monounsaturated fatty acids	One double bond.	Oleic acid (18:1; 9)
Polyunsaturated fatty acids (PUFA)	Two or more double bonds.	Linoleic acid (18:2), Linolenic acid (18:3), Arachidonic acid (20:4)

B. Based on chain length

Group	Carbon length	Examples
Short-chain fatty acids	Less than 6 carbons	Butyric acid
Medium-chain fatty acids	8 to 14 carbons	Caprylic acid, Lauric acid
Long-chain fatty acids	16 to 24 carbons	Palmitic, Stearic, Oleic acids

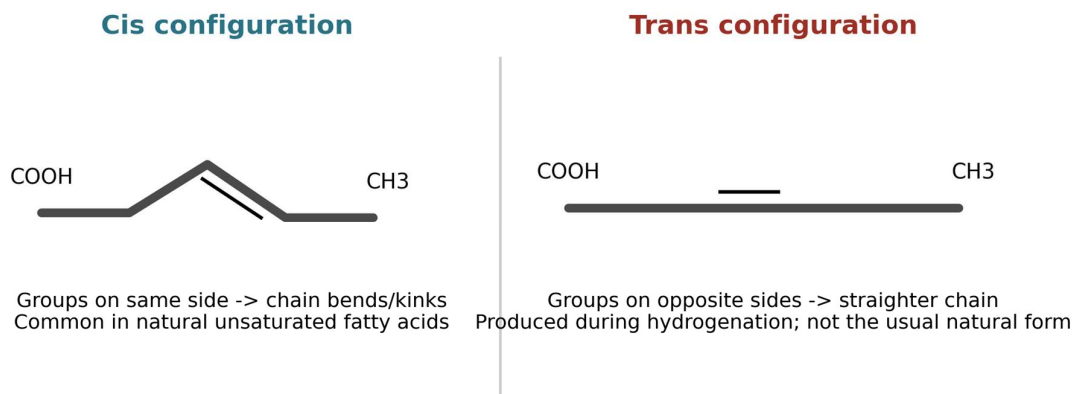
C. Omega series

Series	Meaning	Examples
Omega-3 series	First double bond is 3 carbons from the methyl end.	Linolenic acid (18:3; 9,12,15)

Series	Meaning	Examples
Omega-6 series	First double bond is 6 carbons from the methyl end.	Linoleic acid (18:2; 9,12), Arachidonic acid (20:4)
Omega-9 series	First double bond is 9 carbons from the methyl end.	Oleic acid (18:1; 9)

6. Cis-Trans Isomerism in Unsaturated Fatty Acids

Unsaturated fatty acids show geometric isomerism around double bonds. In the cis form, groups are on the same side of the double bond and the chain bends. In the trans form, groups are on opposite sides and the chain remains straighter. Most naturally occurring unsaturated fatty acids are cis isomers.



Cis and trans arrangement of unsaturated fatty acids redrawn for readability.

7. Essential Fatty Acids

Essential fatty acids are fatty acids that cannot be synthesized by the body and therefore must be supplied in the diet. Chemically, they are polyunsaturated fatty acids.

Fatty acid	Notation	Series	Status
Linoleic acid	18:2; 9,12	Omega-6	Essential
Linolenic acid	18:3; 9,12,15	Omega-3	Essential
Arachidonic acid	20:4; 5,8,11,14	Omega-6	Conditionally essential if linoleic acid is deficient

Biochemical basis for essentiality

Linoleic acid and linolenic acid are essential because humans lack enzymes that introduce double bonds beyond carbons 9 to 10 of a fatty acid chain.

Functions of essential fatty acids

- Maintain membrane structure and function.
- Help in transport of cholesterol and formation of lipoproteins.
- Help prevent fatty liver.
- Serve as precursors for eicosanoids such as prostaglandins, prostacyclins, thromboxanes and leukotrienes.
- Support skin health and wound healing.

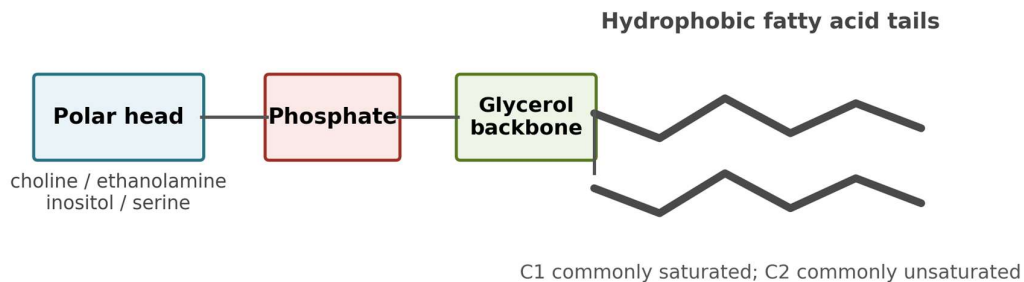
Deficiency of essential fatty acids

Deficiency causes phrynodema or toad skin, characterized by horny eruptions especially on limbs, back and buttocks. It may also cause loss of hair and poor wound healing.

8. Phospholipids: Definition, Types and Importance

Phospholipids are complex or compound lipids containing phosphoric acid in addition to fatty acids, alcohol and often a nitrogenous base. They are amphipathic molecules with a hydrophilic head and hydrophobic fatty acid tails.

General structure of a glycerophospholipid



Phospholipids are amphipathic: hydrophilic head + hydrophobic tails.

General structure of a glycerophospholipid redrawn clearly.

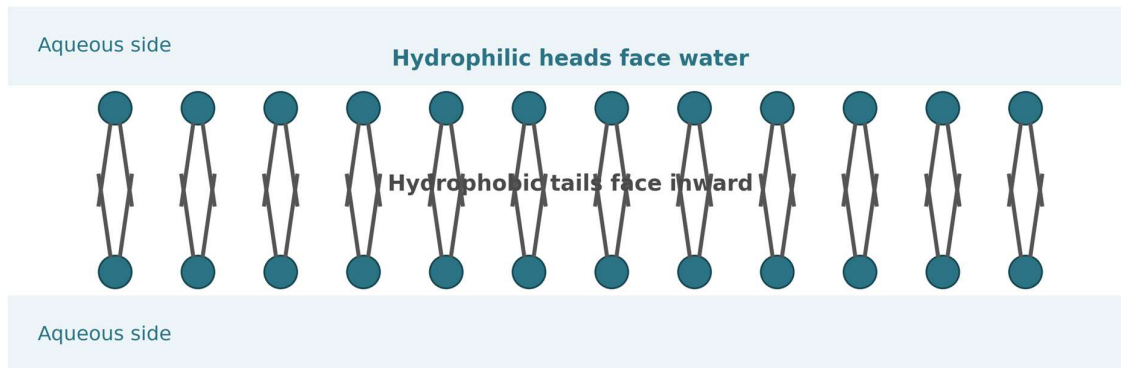
Classification of phospholipids

Type	Key feature	Examples
Glycerophospholipids / phosphoglycerides	Contain glycerol as alcohol. C1 usually has a saturated fatty acid and C2 often has an unsaturated fatty acid.	Lecithin, cephalin, phosphatidylinositol, phosphatidylserine, plasmalogens, cardiolipin
Sphingophospholipids / sphingomyelins	Contain sphingosine instead of glycerol. Important in nervous tissue and myelin.	Sphingomyelin

Phospholipid	Important point
Lecithin / phosphatidylcholine	Most abundant membrane phospholipid; dipalmitoyl lecithin acts as lung surfactant.
Cephalin / phosphatidylethanolamine	Contains ethanolamine; cephalins participate in blood clotting.
Phosphatidylinositol	Important in cell signaling; source of second messengers IP3 and DAG.
Cardiolipin	Important component of inner mitochondrial membrane; required for mitochondrial function.
Sphingomyelin	Major constituent of myelin and nervous tissue.

9. Importance / Functions of Phospholipids

Phospholipid bilayer arrangement in membranes



This arrangement forms the structural basis of biological membranes.

Phospholipid bilayer redrawn clearly to show membrane arrangement.

Function	Explanation
Membrane structure	With proteins, phospholipids form biological membranes and regulate membrane permeability.
Cellular respiration	Lecithin, cephalin and cardiolipin help maintain electron transport chain components in mitochondria.
Fat absorption	Participate in intestinal absorption and transport of dietary lipids.
Lipoprotein synthesis	Required for formation of lipoproteins, helping lipid transport in blood.
Lipotropic action	Help prevent fatty liver by aiding transport of fat away from liver.
Eicosanoid synthesis	Arachidonic acid released from phospholipids acts as precursor of prostaglandins and thromboxanes.
Reverse cholesterol transport	Participate in removal of cholesterol from the body through HDL-related mechanisms.
Surfactant action	Dipalmitoyl phosphatidylcholine lowers surface tension in lungs; deficiency causes respiratory distress syndrome in infants.
Blood clotting	Cephalins take part in coagulation.
Hormone signaling	Phosphatidylinositol produces second messengers involved in action of some hormones.

10. Exam-Oriented Quick Summary

Topic	Must-write point
Definition	Lipids are water-insoluble, organic-solvent-soluble substances related to fatty acids and utilized by cells.
Classification	Simple, complex, derived and miscellaneous lipids. Examples: triacylglycerols, phospholipids, fatty acids, carotenoids.
Fatty acids	Carboxylic acids with hydrocarbon chains; classified by saturation, chain length and essentiality.

Topic	Must-write point
Essential fatty acids	Linoleic acid and linolenic acid; arachidonic acid is conditionally essential.
Phospholipids	Complex lipids containing phosphoric acid; major structural components of cell membranes.
High-yield clinical points	Dipalmitoyl lecithin prevents alveolar collapse; EFA deficiency causes phrynoderma; cardiolipin is important in mitochondria.